

Where You Can Park

Team Members: _____

Every number you have computed this semester came from a sample of something. Today is about how the sample got drawn, and about two ways that goes wrong: an argument you never typed, and a stretch of ground you could not walk to. Open `Class13_Where_You_Can_Park_Skeleton.ipynb`.

Part 1. Two Words That Change the Answer

The marble contest from Lab 05. A bag holds two red, two green, and two blue marbles. You draw three. You win if all three are different colors.

1.1 *Predict first.* Your team's estimate of the probability of winning: _____

1.2 Ten thousand simulated contests, each way.

marble replaced before the next draw: _____

marble kept out of the bag: _____

1.3 Which one matches the contest as described? **REPLACED** **KEPT OUT**

You never told `np.random.choice` which you wanted. Which has it been doing all semester?

Part 2. A Stream With No Name

A bag of marbles is a population you can hold. Now one you cannot: a stream, where the choice is not whether the marble goes back but which stretch of bank you can stand on.

Everything that follows is invented — no such creek, no measurements, numbers straight out of a random number generator. That is on purpose: you can only study a sampling method in a case where the true answer is already known. The notebook holds all 240 hundred-metre reaches along a 24 km stretch of this stream, each with its true chloride concentration. Chloride in real urban streams does come largely from road salt washing off pavement, and that part is not invented. The `access` column says whether a crew could park and walk to the water.

2.1 True mean chloride across all 240 reaches: _____ mg/L

2.2 *Predict first.* Nobody wades 24 km of creek. You park where you can park, and the pull-offs are at road bridges. Before you run anything: a sample drawn only from the easy reaches will be

TOO HIGH **TOO LOW** **ABOUT RIGHT** — because:

2.3 A thousand honest random samples of twelve, against a thousand convenience samples of forty.

	centre of the estimates	SD of the estimates	fraction within 10 of the truth
random, $n = 12$			
convenience, $n = 40$			

2.4 The convenience survey visited more than three times as many sites, and the spread of its estimates is about half as wide. By every measure of precision it is the better survey. Say plainly what is wrong with it, and what would happen if the field crew doubled its effort again.

Discussion

In 1936 the *Literary Digest* mailed ten million ballots and got 2.4 million back, one of the largest polls ever run. It predicted Landon over Roosevelt by about 57 to 43. Roosevelt took roughly 61% of the vote. George Gallup called it correctly with about fifty thousand. Connect this to your answer to 2.4 in one sentence.